# Tunable bank + off-mesh quadrature

## Progress toward the JCP paper

16 October 2026

Numbers from the lanes' generated reports. Slides 1–6 and 8 use the earlier first-order references (provisional); slide 7 uses the new order-verified second-order references.

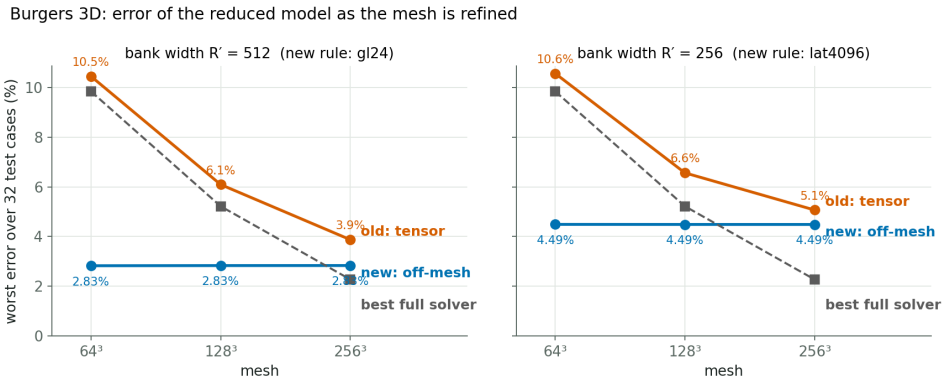
# The idea: evaluate the nonlinear term off the mesh

- **Before:** the reduced model's nonlinear integral $\int \psi\, u(u_x+u_y)$ was computed *on the simulation mesh* (dense sum, fitted EQ, or a giant precomputed tensor). Cost and error both depended on the mesh.
- **Hari's idea:** our bank $u(x) = G(x)c$ is a continuous function. Evaluate it and its *exact* autodiff derivative at a few thousand textbook quadrature points (Gauss, lattice):

$$N(c) = P^{\top}\left[(Bc) \odot (Dc)\right]$$

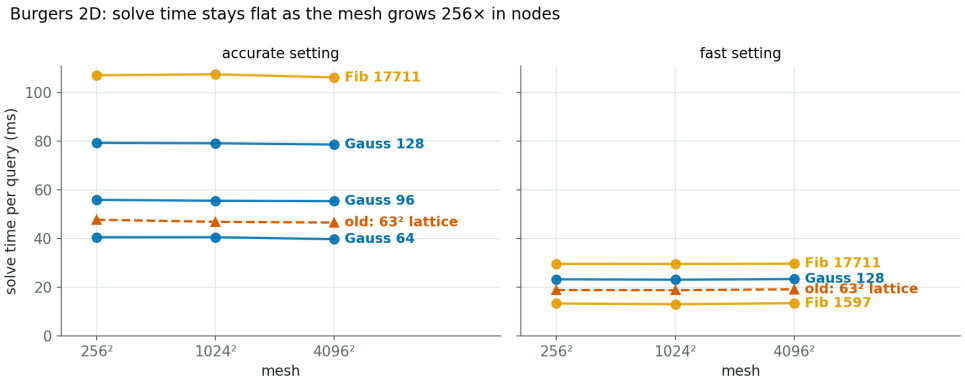
  with $B, D, P$ precomputed once offline.
- No fitting, no per-mesh retraining, and the same points at every mesh.

# 1. Accuracy no longer depends on the mesh (3D Burgers)



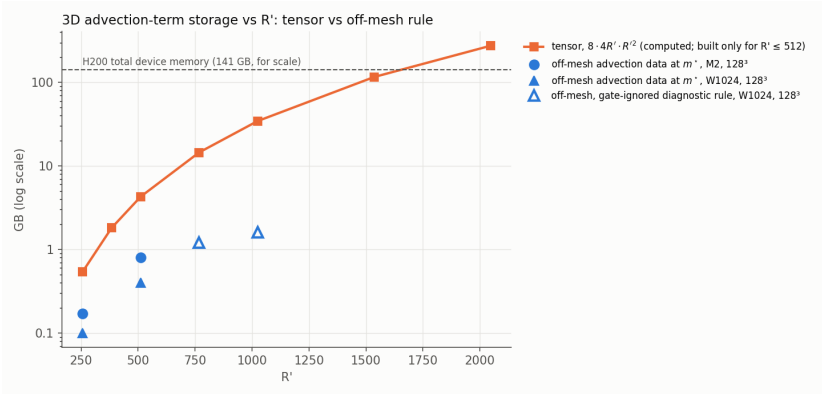

Off-mesh: **2.83 % at every mesh**. The old tensor goes 10.5 → 6.1 → 3.9 %.

# 2. Neither does cost (2D Burgers)



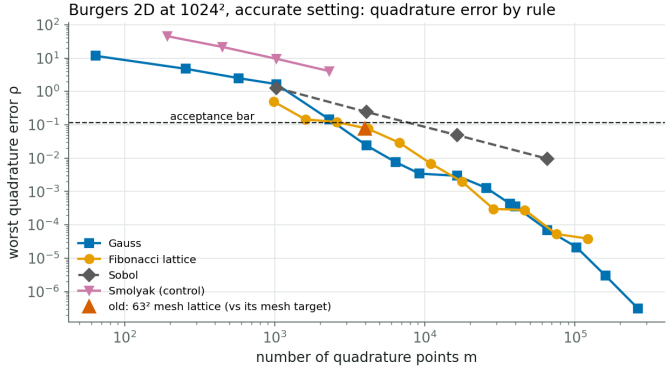

Solve time is identical from $256^2$ to $4096^2$, a mesh $256\times$ larger.

# 3. Memory: wide banks become possible



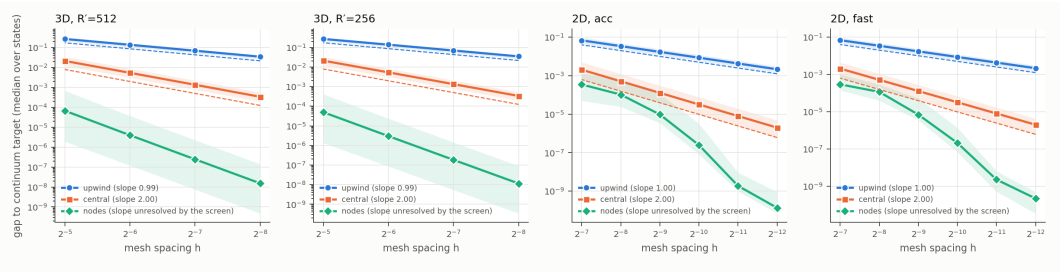

The tensor needs 34 GB at width 1024 and exceeds a whole H200 past about 1600. Off-mesh stays under 2 GB.

# 4. Which quadrature rules work




Gauss and lattices converge fast. Sobol needs about 4× more points; Smolyak fails. Deployed: Gauss (accurate setting), lattice (fast setting).

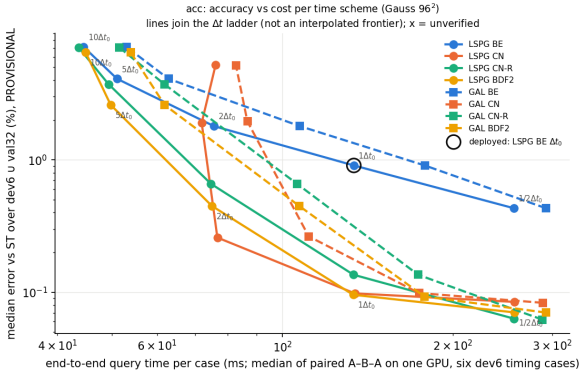
# 5. Why it works: the exact derivative




Bands: 10th–90th percentile over the fixed states. Dashed: C-pos manufactured state (upwind, central). Slopes fitted on the pre-registered window only (the coarsest mesh is excluded); the nodes curve is not a fit.

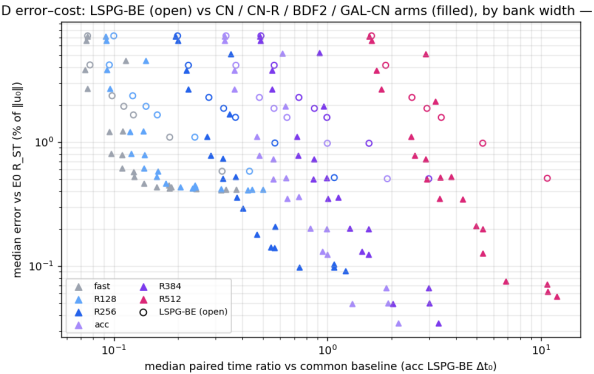
The old error was the mesh's own differencing error: slope 1 for upwind, 2 for central, exactly as theory says. Mesh nodes with the exact derivative reproduce the off-mesh result, so the win is the exact derivative, and quadrature makes it cheap (about 14k points against millions of nodes).

# 6. Next bottleneck, time stepping: fixed with second-order schemes



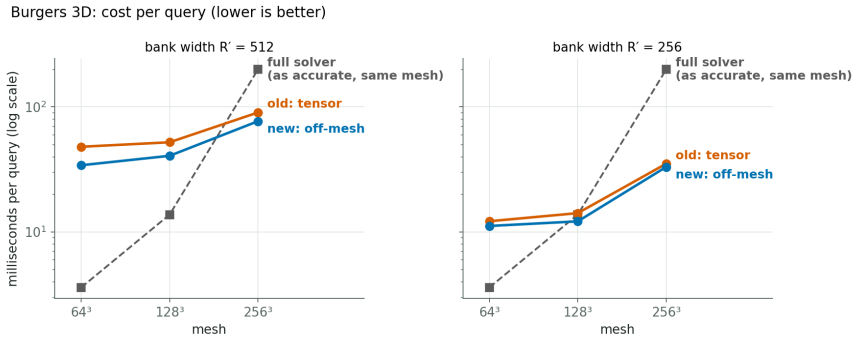

At equal cost, second-order schemes (CN-R, BDF2) have about 9× less error than backward Euler (circled).

# 7. Where we are: against new order-verified references




2D: median error 0.98 $\rightarrow$ 0.20 % at equal cost; about **14**$\times$ faster than a full simulation that is second order in time. 3D: median 1.5 $\rightarrow$ 0.12 % at equal cost.

# 8. Honest limit: 3D speed




We only win at large meshes ($2.6\times$ at $256^3$). Early, unaudited runs show a spectral solver beating us clearly at $64^3$. Large-mesh comparisons ($128^3$–$384^3$) are running now.

# Next steps

- 3D speed lane (running): one-pass solver at big steps, narrower bank, batching, large-mesh comparison against spectral.
- Standard ROM baselines: POD with DEIM, ECSW and GNAT.
- A harder example where spectral solvers struggle: non-box geometry, sharp fronts, non-polynomial nonlinearity.
- Open the sealed test cohorts for the final paper numbers.

**Done this cycle:** order-verified second-order references (3D clean pass; 2D pass, 13 of 204 cases flagged), and second-order stepping combined with the wide bank.